

Crawford Forbes

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Professional Profile

Software Engineer with 10+ years of experience building high-performance web applications, data tools, and frontend systems. From building data-dense tools for enterprise clients like PwC to overseeing multi-team client implementations at Simpleview, I focus on scalable component architecture, performance engineering, and practical technical leadership.

Technical Skills

- **Core Web Technologies:** JavaScript (ES6+), TypeScript, React, Vue.js, Node.js, Modern CSS, HTML5.
- **State & Data Management:** RESTful APIs, GraphQL (Client-Side), Redux, MongoDB, Async Data Pipelines.
- **Build, Performance & Quality:** Vite, Webpack, Lighthouse & Web Vitals Performance Optimization, WCAG 2.1 Accessibility.
- **Architecture & Delivery:** AI-Augmented SDLC (LLM Workflows), Technical Discovery & Scoping, Cross-Functional Technical Documentation, Code Reviews & Mentorship, CMS Architecture (WordPress, Custom).

Professional Experience

Simpleview (a Granicus company) | *Software Engineer/Team Lead* | Jul 2018 – Aug 2025

- **[Performance Engineering]** Architected a complete rewrite of the **core booking engine in Vue.js**, consolidating 12 disparate API calls into a **single optimized request** and implementing custom **hydration logic** to ensure seamless UI transitions on load.
- **[Technical Discovery]** Led the **technical discovery and scoping process**, evaluating design prototypes for feasibility and providing **architectural solutions** to keep features within budget and timeline constraints.
- **[Data Orchestration]** Engineered a data-driven Listings widget for Visit Bellevue, utilizing **GraphQL** to consume proprietary CRM data and integrating **third-party REST APIs** to inject **Yelp reviews** and **real-time booking links**.
- **[Process Optimization]** Drove a **~30% reduction in team build times** by analyzing development workflows and implementing **automation scripts for data migration** and CRM form submission.
- **[Large-Scale Delivery]** Directed **multi-team engineering efforts** (e.g., Travel Wyoming), managing feature prioritization and code quality for up to **eight contributors** across dozens of custom components.
- **[Performance & Accessibility]** Optimized front-end performance and **accessibility (WCAG)** across all builds, utilizing **Lighthouse and WAVE** to ensure high-speed delivery and cross-browser compatibility.
- **[Technical Documentation]** Standardized engineering excellence by authoring **internal technical manuals**, reducing the bus factor and streamlining onboarding for new developers.

Great Scott Design | *Full-Stack Developer* | Aug 2017 – Jul 2018

- **[Full-Stack Ownership]** Architected and maintained high-performance **WordPress and WooCommerce** sites, managing the full project lifecycle from initial stakeholder discovery to final production deployment.
- **[Security Operations]** Implemented a comprehensive **system-hardening strategy**, including real-time monitoring and manual **database restoration protocols** to maintain site integrity and high-availability standards.
- **[Technical Consulting]** Provided daily technical consulting for clients, translating experience-based insights into **actionable UX improvements** and modern web features.

Elegant Thought | *Front-End Developer* | Sep 2015 – Jun 2017

- **[Technical Leadership]** Served as **Lead Frontend Developer** for the **PwC MoneyTree Explorer**, a secure data exploration tool built to visualize **complex, multi-dimensional financial datasets**.
- **[Enterprise Architecture]** Architected and developed **high-performance SPAs** for global enterprise clients,

including **PwC**, using **React** and **Redux** to meet rigorous security and performance standards.

- **[State Management]** Engineered a robust state architecture using **Redux and Redux-Thunk** to manage deep-dive filtering and **real-time interactive charting** across multiple levels of granularity.
- **[CMS Integration]** Seamlessly integrated **React-based clients into Adobe Experience Manager (AEM)**, navigating the strict security, branding, and technical restrictions of a **global enterprise ecosystem**.
- **[Data Visualization]** Developed interactive visualization suites using **Highcharts and Highmaps**, transforming raw backend data into dynamic, accessible **maps and charting systems**.

General Assembly | *Web Development Immersive Instructor* | Nov 2014 – Jun 2015

- **[Mentorship & Leadership]** Technical Lead Instructor for two 12-week full-stack immersive cohorts, mentoring **50+ adult learners** through the transition into junior software engineering roles.
- **[Code Quality & Review]** Facilitated **rigorous code reviews** for student-led applications, identifying architectural anti-patterns and enforcing **clean code standards**.
- **[Subject Matter Expertise]** Instructional Lead for complex systems communication, including **RESTful API design, MVC frameworks**, and real-time data transmission via **WebSockets**.

Technical Research & Independent Projects

[Pelagos-7 & Portfolio](#) | *Lead Developer* | Oct 2025 – Present

- **[Architecture & AI Orchestration]** Architected a high-performance generative music SPA using **React 19, TypeScript, and Zustand**, leveraging **agentic workflows** for automated GitHub ticket generation, unit testing via **Vitest**, and synchronized commit/PR messaging to rapidly iterate on complex Audio/GSAP logic.
- **[Audio Engineering]** Developed a centralized **AudioEngine using Tone.js** for polyphony management and **look-ahead scheduling**, ensuring precise timing and low-latency playback through a custom-built, musically synchronized timing system.
- **[Tooling & Automation]** Engineered **automated media pipelines** using custom **Node.js scripts** to handle video transcoding and image optimization for performant, responsive delivery.
- **[Systems Design & Discovery]** Utilized LLMs for **high-level technical discovery**, evaluating architectural trade-offs between library selections and state patterns to ensure a performant, scalable, and maintainable codebase.
- **[Stability & Resilience]** Developed a robust error-handling architecture leveraging **React Error Boundaries** and custom utility modules to ensure graceful degradation; managed third-party dependency risks using **patch-package** to maintain custom library fixes safely.
- **[Accessibility & SEO]** Optimized application **SEO** and discoverability through automated sitemap management and **accessible UI patterns (WCAG 2.1)**.

Sunshine Nights | *Lead Developer* | Jun 2015 (initial build); 2016 – Present (maintenance & updates)

- **[Full-Stack Security]** Architected and maintained a custom, secure CMS using **Ruby and Sinatra**, implementing user authentication, session expirations, and protected RESTful endpoints to manage dynamic content.
- **[State Management]** Engineered a sophisticated **React/Redux SPA** featuring a custom container architecture and **predictable state management** for complex UI overlays and album promotion modules.
- **[Systems Engineering]** Developed a **custom state management system** to coordinate asynchronous data fetching, loading transitions, and global UI states across multiple **REST endpoints**.
- **[Performance Optimization]** Transitioned core frontend components from React to a **lightweight Vanilla JS and AJAX (XMLHttpRequest) architecture** to optimize load times while retaining React for high-interaction features.

Education

Web Development Immersive | *General Assembly*, New York, NY | Aug 2014 – Oct 2014

B.A. in Music | *Susquehanna University*, Selinsgrove, PA